

Day 7 – Shell Scripting: Process Management, Background Jobs & Cron

Objective:

Learn how to monitor running processes, manage background jobs, schedule tasks with cron, and build disk monitoring scripts.

1. Process Monitoring:

ps -ef : Show running processes

pgrep process : Find process PID

Example:

pgrep nginx

2. Killing Processes:

kill PID : Stop process gracefully

kill -9 PID : Force kill process

3. Background Jobs:

Run command in background using &

Example: sleep 60 &

Get background process PID using:

\$!

Example:

sleep 60 &

echo \$!

4. Cron Jobs (Task Scheduling):

Edit cron table: crontab -e

Cron Format:

* * * * * command

Minute Hour Day Month Weekday

Example Cron Jobs:

Run every minute:

```
* * * * * /path/script.sh
```

Run daily at 2 AM:

```
0 2 * * * /path/backup.sh
```

5. Disk Monitoring Script Logic:

```
df -P | tail -n +2 | while read line
```

Extract usage column (\$5) and partition (\$6)

Remove % using `tr -d '%'`

Compare usage with threshold

Example Logic:

```
USAGE=$(df -P / | awk 'NR==2 {print $5}' | tr -d '%')
```

Best Practices:

- Use `df -P` for consistent parsing
- Always quote variables
- Validate arguments
- Avoid parsing errors from command output
- Use cron for automation

Day 7 completed successfully